

Archaeological Analysis of Stable Isotopes

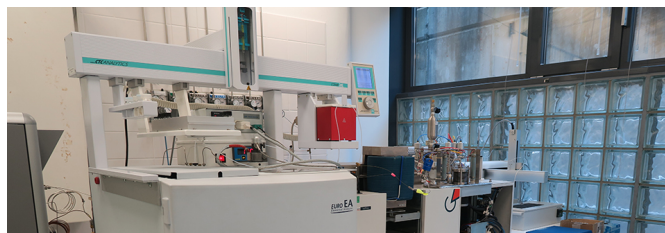


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The most common circumstance under which stable-isotope ratios of carbon and nitrogen are used in archaeological analyses is when archaeologists want to determine the diet of animals, especially humans, in the past. Animal bones contain collagen, a protein with both nitrogen and carbon that is often well preserved in variety of archaeological contexts, since it is protected by the durable inorganic fraction of the bone, hydroxyapatite. Hydroxyapatite can also be analyzed for carbon isotopes, but not nitrogen isotopes, as it lacks nitrogen in its chemical structure. Thus, most often, archaeologists analyze collagen to retrieve both carbon and nitrogen values at the same time.

The process begins with taking a bone sample and extracting collagen from the bone. This is a destructive process, so the use of it on human remains in particular is only possible under certain circumstances. Chemically, however, the process is relatively simple, as collagen is easy to isolate chemically. The process begins by grinding a small piece of bone into a powder and decalcifying it with an acid, which removes the inorganic component of the bone. This is relatively simple chemistry that can be done quickly. Once the collagen is isolated, it is weighed and sometimes directly examined using spectroscopy to ensure that it is present in expected quantities and chemically intact. Following these quality-control steps, the collagen is analyzed using a form of mass spectrometry known as **isotope ratio mass spectrometry**, or IRMS. The IRMS will produce mass ratios of heavy to light carbon and nitrogen stable isotopes for the sample. These values can then be compared with those of the standard to produce $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values for the sample.

Isotope ratio mass spectrometry, abbreviated IRMS, is a form of mass spectrometry that is preferred for the analysis of stable carbon and nitrogen isotopes.





An isotope-ratio mass spectrometer.

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